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SVF-CDM: Sky View Factor, a Continuous Digital Model for large cities using SVFPy

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ABSTRACT

The Sky View Factor (SVF) has been used as an index of dimensionality and complexity reduction in several areas of knowledge to model and qualify the urban environment and the landscape. This article proposes a new method to calculate the Digital Continuous Model of SVF (SVF-CDM) from LiDAR 3D ALS surveys for large cities. The method is described from the perspective of single-point computing and then adapted to a more efficient heuristic computational process called here SVFPy . The SVF-CDM processing was carried out for the entire length of the São Paulo and New York cities, with a spatial resolution of 50 cm and 1.25 ft (38.1 cm), respectively. Method validation was based on comparing a sample of random points, and the median errors were -0.53 (São Paulo) and 1.37 (New York) percentage points. The results indicate that the SVF-CDM has advantages over other digital models and features for urban morphology analysis. Whether visually or through its dimensional and statistical expression, the presented model manages to express the nuances and morphological aspects of the studied areas. In this way, we conclude that we are facing a significant advance for the models of urban studies, not only because it considers the perspective of the pedestrian but also because it is a powerful resource for studies and disciplines related to the urban form without disregarding all its complexity.

Contents

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1. Introduction

Sky View Factor (SVF) is a way of reducing the dimensionality of the urban form [1] from the amount of sky that can be seen without any obstruction [2,3], from a certain point of view. In cities, urban morphology and their complexity are often represented by multidimensional models [4–7], and the SVF can play an important role by enabling a representation of the relationship between built elements, open space systems and the environment, from the perspective of people in the urban fabric.

However, to the best of authors knowledge, there is still no methodological approach that allows calculating the SVF for large cities or massive extensions of land through a continuous digital model of SVF with sub-meter precision, for several scales of studies. The present work proposes a method to compute this continuous digital model, that is a raster matrix representation with all points calculated with the sky view factor, throughout the cities, without interruptions or gaps. This model may serve as an input for several researches, as SVF has been considered as a determining factor or correlated to different phenomena within city sciences.

The objective of this article, therefore, is to propose a method for calculating the continuous SVF for large urban areas such as the city of São Paulo and New York for submetric spatial resolution. And so, to provide a Continuous Digital Model of the Sky Vision Factor (SVF-CDM), that allows the study and analysis by several disciplines and urban areas such as: public health, micro-climate, urban ecology, thermal and environmental comfort, transport that depend on GPS and urban legislation for land use and occupation. Above all, for this work, the focus will be on the idea that the SVF-CDM is a model capable of expressing views of the complexity of urban form adaptable to different scales of urban studies. In a broader view, it is expected to disseminate and encourage the use of the SVF as an urban indicator for urban planning, research and formulation of public policies on land use and occupation, transport, health and the environment.

This work is organized as follows. In addition to this introduction, we present the review of the literature, materials and methods, results, conclusions and future work. The literature review emphasizes the uses and applications of the SVF in urban research, as well as in the previous methods approaches already proposed for obtaining digital models of SVF. The proposed methodology ranges from the theoretical calculation of the SVF, proposed by the literature since its inception, to the development and validation of the heuristic that allows, in an unprecedented way, the calculation of an SVF-CDM for large cities. To do so we apply this processing to the cities of São Paulo and New York. The results are presented in order to emphasize the potential use of the SVF-CDM as a dimension of the complexity of urban form at different scales in a simple way through its descriptive statistics and images. The final considerations describes possibilities of uses, future work, possible methodological improvements for more specific needs and that may require greater precision, based on the legacy of this new possibility of calculating SVF-CDM for large urban areas.

2. Literature review

From the moment that [2] described the relationship between the SVF and the urban microclimate, several other specialties raised hypotheses of the correlation between the SVF and their objects of study. Many of them are mainly motivated by the fact that the SVF captures an important dimension of the complexity of the urban form. On the other hand, since [3] proposed a graphical estimate for calculating the sky view factor, several methodological proposals have been described in the literature, although none of them have been successful in calculating a sub-metric continuous model for large cities. We divide this literature review section into two sub-sections, one with possibilities of use and the other discussing previous approaches for the calculation of the SVF.

2.1. SVF application possibilities

There is a lack of articles that relate SVF directly to urban morphology, for instance, identifying building patterns and scenarios. However, for all the specific areas described in this review, the justification for using the SVF as part of the proposed models is that it captures an important dimension of the complexity of the city and the diversity of the urban form. Therefore, the SVF has the potential to serve as an index for studies of models that simulate changes in land occupation and morphology [8] and thus be part of the framework of urban indicators for urban planning and their metrics.

Urban morphology has been increasingly mapped in order to better capture climate conditions in urban environment, considering the potential impacts of climate change. Stewart and Oke [9] study the Local Climate Zones (LCZ), which aims to detail the climate responses of different urban structures based in a SVF model and its relation with different urban morphology characteristics, such as density, percentage of impermeable area, roughness, coating materials, among others.

The concept of Local Climate Zones (LCZs) is closely related to the Sky View Factor (SVF), as the morphological and structural characteristics of different local climate zones directly influence the amount of visible sky from ground level. LCZs categorize urban and rural areas based on features such as building height, vegetation density, and land cover. In densely built urban areas (LCZ 1 to LCZ 3, for example), the SVF tends to be reduced due to the presence of tall buildings and narrow streets, which limit sky visibility [9]. In contrast, in less urbanized or open areas (LCZ 6 to LCZ 10), the SVF is higher, reflecting greater sky exposure due to less obstruction from buildings and vegetation. Thus, the study of SVF in different LCZs can provide valuable insights into the urban microclimate and the thermal comfort of inhabitants [10].

Also it is possible to investigate the SVF relation from SVF with surface energy balance, air circulation and outdoor thermal comfort [11]. Some researches explore the capacity of using SVF as a proxy to predict heat islands with distinct outcomes, while [12] succeeded in getting heat island from SVF, Minella et al. [13] claim that the SVF had a low correlation with the microclimate, although it was correlated with other thermal comfort indices, such as PMV (Predicted Mean Vote) and PET (Physiological Equivalent Temperature). For [14] SVF is an indicator to evaluate the performance of both the human perception of the built environment and the microclimate.

Concerning the human perception, some studies relate mental health to urban parameters, including the use of SVF [15–17]. Wang et al. [18] hypothesized that denser neighborhoods with a low percentage of sky visibility, measured from Street view images, may be associated with a greater probability of depression and anxiety in the elderly, however, it does so directly, attributing a causal relationship between sky view and walkability without presenting much evidence of this parity.

Besides the aforementioned relation of the SVF with urban quality of life, studies have suggested a strong correlation with different pollution dimensions, like air and noise. Yilmaz et al. [19] observed that air pollution is less dense in regions where buildings are more isolated, that is, with a greater proportion of visible sky. Other studies that propose models of pollution in urban environments consider the SVF as an essential dimension and relate higher values of SVF to worse conditions of atmospheric pollution [20,20]. Regarding noise propagation [21]. Silva et al. [22] conducted a study in the Portuguese city of Braga and concluded that noise propagation was inversely proportional to the calculated SVF, i.e., the highest noise levels were obtained where the sky was most obstructed by buildings. The authors consider that the SVF has the potential to be used in research and models of noise propagation in urban environments.

To sum up cities have been installing a growing network of different sensors to support data collection searching for strategies to monitor the urban environment looking forward more sustainable cities. In many cases the sensors depend on the GNSS systems. Accuracy in location and speed of data acquisition from GPS are important for this sensors, therefore, the mapping of possible areas with a lower probability of being targeted with the satellite constellation can prevent the inaccuracy of these systems and even encourage other strategies for locating [23–25], especially considering that the target with the constellation of satellites is a direct function of SVF.

So SVF has been widely use in urban applications, and different approaches are explored as discussed in the next section.

Robinson [26] concluded in his study that, when comparing simulations of the Sky View Factor (SVF) with those of solar irradiation, that the absence of considerations for direct solar radiation, sky anisotropy, or reflected radiation significantly impairs the ability of SVF-related indicators to provide accurate insights into the potential for natural radiation. However, the aggregated SVF correlates reasonably well with irradiation, allowing for analysis about overall radiation availability to be made with some degree of confidence. Therefore, while the use of SVF is valuable, it should not be employed in isolation and should be complemented by other studies and simulations for a more robust assessment.

2.2. Previous approaches

As evidence grows that the sky view factor affects important urban aspects as seen previously, it becomes more necessary to develop methods and techniques to quantify, study and compare the proportion of visible sky. Evidence of this are the recent publications of articles proposing techniques for calculating the SVF. Miao et al. [27] classified the methods into the following categories: geometric; “of fisheye cameras”; the “GPS methods”; simulation based on surface models and 3D models of cities; and the approaches that the author calls Big Data from Street View images. Although the proposed classification is interesting, it does not consider methods based on 3D LiDAR surveys, and the authors justify this choice by pointing that they are not present in many cities. But as technologies advance there is a promising scenario of new 3D LiDAR surveys around the world. Kidd and Chapman [28] consider that the calculation of continuous SVF fits in simulation models from Digital Surface Models (SDM). In practice many methods end up fitting into two or more classifications and depend on the resources and data available, mainly because calculation of the SVF depends on the geometric relationship of the observer’s point with a theoretical dome, as proposed by Johnson and Watson [29], when he described the first theoretical model for calculating the SVF and its graphical estimate [3].

Many articles were identified in the literature using techniques from Google Street View images [8,18,30–34], whose advantage is the availability of data in urban areas of the planet, in addition to the use of automation techniques and computer vision in the segmentation or differentiation of images. However, they end up being limited to discrete samples at specific points, which makes a broader, continuous and systemic view impossible. Middel et al. [1] also used Street View images to create an optimized routing system for thermal comfort in the city of Arizona, USA, whose validation is done in an interesting way: a 3D reconstruction from the automated flyover in Google Earth.

Kidd and Chapman [28] conducted a survey in south Cornwall (UK) an area of 4 km2 calculating the SVF-CDM, using a method from a 3D LiDAR model for the calculation from an DSM. The authors compared with calculations performed from photographs and concluded that calculating SVF-CDM that both techniques reached similar results with the advantage of SVF-CDM being a continuous model.The authors calculated the SVF value every 2 meters from the Digital Surface Model for a maximum distance of 1000 m. In other words, the authors did not use a heuristic and did not worry about optimizing their calculation method.

In addition to the authors, other authors also use 3D LiDAR datasets, but as the primary source to process the Digital Surface Model (DSM) which is then used for the actual processing [12,35,36].

It is important to point out that none of the studied articles presented results in large areas such as a city like São Paulo (1524 km2) or New York (778 km2) and with spatial resolution in the continuous scale below 1 meter per pixel. Gál et al. [37] calculated the continuous SVF with spatial resolution of 5 m for the urbanized region for the city of Szeged, Hungary and [38] calculated for the city of London, UK using a 4 m spatial resolution SDM.

In the case of Hungary, to calculate the SVF spatial resolution of 5 m, the authors made a grid with the pre-defined distance and then for each of the points they calculated the Sky View Factor considering angles and taking the largest angle for each time zone, without an optimization process of this algorithm. For the London study, the authors adopted the shadow projection method. This method consists of calculating the value of the shadows generated by the angle of sunlight from various positions and then weighting each calculation to determine the sky view factor.

Hodul et al. [39] proposed a method for calculating the SVF on a global scale, however, using Landsat images with a spatial resolution that does not support the pedestrian level. Computational performance is addressed only by Li and Wang [40] who used dataset processing for the city of Philadelphia (369.59 km2), USA, with a spatial resolution of 1 m in 20 min using a computer running Ubuntu Linux system with 16G of memory, Intel Core(TM) i3-4170 processor, and NVIDIA GeForce GTX 1060 6 GB GPU graphics card. The authors’ methods did not include a multi-resolution approach, they used 2000-pixel grids to calculate the SVF and added a 150-meter border to the entire model to avoid a very abrupt drop.

3. Materials and methods

We present the proposed method in two parts. Firstly, defining the calculation used to determine the visible sky factor, already discussed in the literature. Then proposing a more computationally efficient method. Finally, the result of the proposed method is compared with a random sample to evaluate it.

3.1. ALS LiDAR 3D data

Several cities around the world have ALS LiDAR 3D surveys available, we chose two of the most populous and extensive cities, São Paulo and New York, with different morphological and spatial characteristics to be able to explore the proposed method to different scenarios. Although [28] have demonstrated and proposed a technique for calculating the SVF directly from the 3D LiDAR survey to enable an extensive, continuous and complete dataset of the cities, it was decided to use a Digital Surface Model based on the existing 3D LiDAR survey, detailed further in the method application section. But what does not prevent the use of this method to process the SVF directly in Digital Terrain and Surface Models.

3.2. The classical SVF calculation

The SVF, which is the proportion of sky visible from an observation point, can be calculated by dividing an imaginary, semi-spherical dome over the observer, whose base is the plane perpendicular to the zenith axis, into zones (k) with angle θ of the same opening [41], where each fuse is divided into two transverse sections, perpendicular to the zenith axis, by a meridional plane parallel to the base plane XY. Thus, calculating the area of the visible part over the total area of the time zone, the sky view factor of the fuse in question k is obtained, as shown in the equations below. The proportional sum of the calculated factor for each of the fuses is the Sky View Factor of that observation point. This process is partially illustrated in Fig. 1.

Considering A_{sphere} as the area of the sphere, $A_{sphere} = 4\pi r^2$ it is possible to define the area A_{dome} of the hemispherical dome as:

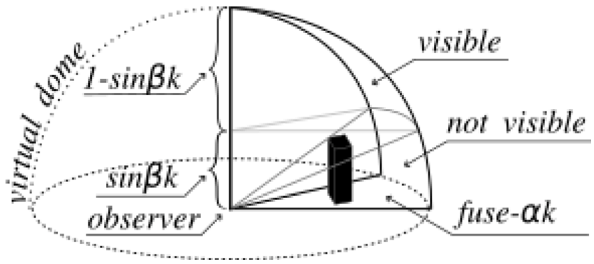

Fig. 1. Schematic diagram of the theoretical method for calculating the SVF.

$$A_{dome} = \frac{4\pi r^2}{2}$$

The sphere can then be divided into fuses with angle α, where $\alpha = \frac{2\pi}{n_{fuses}}$

The area of each zone is known and given by $A_{fuse} = \frac{2\pi r^2 \alpha}{2\pi}$, therefore $A_{fuse} = r^2\alpha$

The zone then can be split into A_{vs} and $A_{shadow} = A_{dome} - A_{vs}$. Or even directly calculating the ratio of the visible sky:

$$R_{svf} = \frac{A_{vs}}{A_{dome}}$$

Knowing the area of the fuse A_{fuse} and the maximum height of the obstacle of vision of that time zone calculated by the angle, β we can then subtract the area of the fuse from the visible sky area A_{vsfuse}, being $A_{vsfuse} = r(r - \sin\beta)\alpha$

Therefore, for each fuse the Sky View Factor is given by

$$R_{svffuse} = \frac{r(r-\sin\beta)\alpha}{r^2\alpha} \frac{r-\sin\beta}{r}$$

Therefore, the Sky View Factor for a given point is given by:

$$R_{svf} = \frac{1}{N_{fuses}} \sum_{k=1}^{N_{fuses}} \left(\frac{r - \sin\beta_k}{r} \right)$$

As we intend to obtain a sky view factor, the radius r can be replaced by any positive number greater than zero.

$$R_{svf} = \frac{1}{N_{fuses}} \sum_{k=1}^{N_{fuses}} \left(1 - \sin\beta_k\right)$$

For a discrete calculation, it is possible to divide the imaginary dome (the semi-sphere) into zones θ and angles ϕ into portions with the same sine value. That is, differences of equal values of the inverse of Sine ($\sin^{-1}(x)$), ArcSine ($ArcSin(x)$), produce equal areas of caps.

3.3. Computational implementation: SVFPy

The theoretical method presented, if implemented literally, would depend on the relationship of each point with each point of the SDM, becoming a problem of high computational complexity $O(n^2)$ in the worst case. Especially in a dataset with $6e+09$ of points, $3.7e+19$ computational interactions would be required, which would make it impossible to process the dataset completely. In this way, it was decided to develop a heuristic method capable of expressing the relationship of the points more efficiently and still preserving accuracy.

Thus, successive strategies were chosen to optimize computational resources: A discretization of observation angles, B generalization of spatial resolution for each vertical angle, C processing in 'kernels' arrays, based on the parameters described in the 1 table.

A As seen previously in the calculations described in the literature, the fuses of the imaginary dome are discreetly divided, and the angle of the observer with the obstacle is calculated to then determine the portion of the fuse with a view to the sky. To obtain computational performance, the vertical angles ϕ were divided, from the appropriate parameter. So that each angle represents a fraction of the same area of the dome, thus, the semi-sphere can be divided by varying the parameters of the quantity of fuses and the number of vertical angles. That when multiplied result we have the result accuracy of the calculated result. For example, for the two cities processed, this product results in 256, an 8-bit integer that guided the choice of such values, in addition to representing a balance between the vertical and horizontal portion of the factor, with 32 zones θ and 8 vertical angles ϕ.

B Once the vertical angles are defined, we have a relationship between the pixel projection and a horizontal plane perpendicular to the observer and the obstacle defined by the tangent of ϕ. That is, for each variation on the Z axis there is a variation of $\tan(\phi)$, and as the intention is to calculate the variation of Z in relation to the plane, XY it becomes possible to optimize the sets of data to be observed. Thus, it was decided to generalize the SDM to require fewer calculations per processed kernel, forming a pyramid of SDMs that are loaded from each calculated vertical angle ϕ. For example, for an angle, ϕ, the next integer value is calculated and a new upscaled dataset is calculated. The generalization calculation uses the statistical mode of the values to assign to the data set with a higher spatial resolution, that is, more generic, considering the roughness of the built urban environment and also the generalization literature from LiDAR 3D [42]

A computationally costly process observed was the reading of the dataset and memory allocation. C Looking to reduce this cost, it was decided to process the dataset from kernels, using the matrix calculation resources of the NumPy library [43]. The definition of the matrix size is given by the Kernel Side Size parameter ($KernelSizeSide$) multiplied by $\sqrt{2}$, added to a Pad calculated from the parameter defined by the Maximum Height Difference ($MaxVerticalDifference$), which depends on the angle ϕ, so that obstacles outside the processed kernel can be represented in the calculation whatever the angle of rotation of the fuse.

$$Pad = MaxVerticalDifference \times \tan(\phi)$$

$$KernelSizePadded = ((\frac{KernelSizeSide}{2} \times \sqrt{2}) + ceil(Pad)) \times 2$$

To process the data then, the complete matrix of the SDM is partitioned, in squared kernels defined by $KernelSizeSide$, considering that for each angle ϕ there is only one resolution already previously processed (B). Then, the Matrix loaded in memory is rotated[1,2] for each angle θ and its projection in XY is calculated, multiplying its value, Z by the tangent of ϕ. After obtaining the values of X (Z projected into X) and adding their distance to the first element, the list is sorted and the maximum accumulated value is computed for each element. In this way, it is possible to compare the previous list and test whether such value is blocked by another value projected further ahead, as illustrated in Fig. 2. In the Figure, it is possible to see that the order varies according to the angle and thus, in a more computationally efficient way, it is possible to determine if a certain element has a blocked relation with the sky just by comparing it with the original order.

The Sky View Factor (SVF) is calculated from the count of all false values for blocking divided by the total number of interactions ($N_\theta \times N_\phi$). Thus, the sum of these binary sub-products can be easily

[1] The Matrix is not rotated in memory, only a pointer to the indices is redone at each iteration.

[2] The rotations are not actually performed at all once the transposition of lines in columns and the inverse order can represent the orthogonal angles and thus saving more resources.

Table 1
Method input parameters.

Parameter	Description	Assigned values	
		São Paulo	New York
Number of fuses	Number of parts or horizontal angles the dome will be divided into	32	32
Number of vertical angles	Number of parts or vertical angles that the dome will be divided	8	8
Kernel side size (pixels)	Height in pixels of square equilateral kernel for processing batch	2500	2500
Observer height (m)	Altura em metros do observador	0	0
Max distance (m)	Maximum horizontal distance to consider an obstacle	10 000	10 000
Maximum height difference (m)	Maximum height difference between the observer and the obstacle to calculate the 'pad' added to the Kernel sides	180	600

accounted for derivations such as calculations of specific angles such as the view of the Sun or another celestial element.

The complete method written in Python is available in the project's repository.

3.4. Method application

For the city of New York, the 3D LiDAR aerial survey carried out in 2021, already classified, according to ASPRS criteria, was used. A DSM (Digital Surface Model) was then processed with a spatial resolution of 1.25ft. (38.1 cm) For the city of São Paulo, a LiDAR 3D aerial survey carried out in 2017 was used to process a DSM with a spatial resolution of 50 cm. The use of different measurement units was deliberate to be able to validate two different CRS.

Both data processing, New York and São Paulo, were performed on a personal machine with an 8th generation i5 processor and 32 Gb of memory. Each data set took about 12 h from pre-processing to the final set ready for analysis. This is a satisfactory result considering the physical dimension of the two cities studied.[3]

3.5. Method evaluation

Considering the size of the datasets for São Paulo[4] and New York,[5] a confidence level of 95%, with a tolerable margin of error of 5%, in both cases samples were calculated with 385 random points with a minimum distance of 100 m from the edge of the boundaries and 500 m between the samples, whose calculations will be available in annexed to this paper. For each sample, the SVF was calculated considering the interaction of the observer's point of view with all the other points. Although computationally inefficient, this method is the standard to validate the heuristic used. Based on calculations, such data were compared with those obtained by the method proposed here, the SVFPy. Then, the difference in percentage points between the two values obtained for each of the samples in each of the cities was calculated. The median error obtained for the São Paulo data was −0.53% with a standard deviation of 0.09. As for the New York data, the median error calculated for the sample was 1.37% with a standard deviation of 0.09.

Therefore, the result of the validation is quite positive. The expected error could be up to 6% in the worst case, considering the parameter used of 8 discrete angles ϕ for calculation. However, the interaction between the arrangement possibilities of multiple scales and spindle angles (θ) turns the average error into a range that does not interfere with the SVF analysis. In applications that require more precision, different parameter values can further improve accuracy.

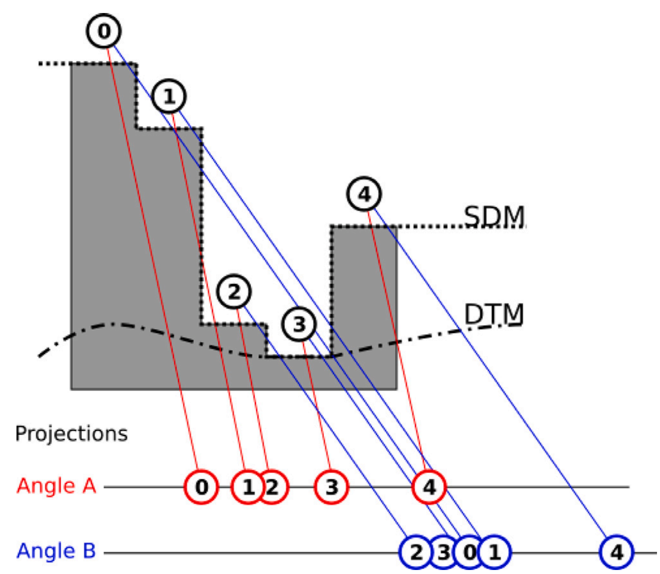

Fig. 2. Schematic diagram of part of the heuristic method implemented SVFPy detailing the calculation of the verification of the presence of obstacles for each angle ϕ by ordering the projections of Z in X.

3.6. Method limitations

SVFPy uses the SDM as input, being a simplification of something solid, in such a way, it disregards buildings with free spans and marquises, as well as openings in the vertical faces of the buildings.

The data are restricted to the administrative limits of the cities, sometimes exceeding a few hundred meters, making it impossible to refer to and calculate the view of built or natural elements outside the limits of the municipality. In practice, knowing that there is no representative geographic feature in the surroundings of the two cities, it is possible to state that this will have little effect on the two data processing presented. Still, it is necessary to make these considerations when the analysis focuses on closer locations to the borders of cities. It is possible to use other resources such as the Shuttle Radar Topography Mission (SRTM) mission carried out in 2000 with a spatial resolution of 30 m to compose the model beyond the limits of the studied area.[6]

[3] 13,679,684,656 (y: 144,826 ex: 94,456) pixels for São Paulo and 15,121,391,628 (y: 122,377, x: 123,564).

[4] The administrative limit used for São Paulo has a calculated area of 1,527,000,000m2, considering a pixel of 50 cm, the calculated result has 6,108,000,000.

[5] O The administrative limit used for the city of New York has 822,920,000m2, considering a pixel of 1.25 ft (38.1 cm), the calculated result has 3,374,835,958 SVF measurements.

Fig. 3. Result of calculated SVF-CDM portions on the left, SDM in the middle and aerial photos on the right, with a portion of New York City above and São Paulo below.

There is also a temporary limitation in the method due to the use of flat projection models (SIRGAS 2000/UTM 23S and NAD83(2011)/New York Long Island (ftUS)) that disregard the Earth's curvature in this current version of the method. This limitation should influence more over areas with a vast view and full sky factor than in regions where the sky factor is more limited, with obstacles limited to the vertical angle. However, this research intends to update the model to work with the Earth's curvature and enable its use on even broader scales. It is necessary to reflect on the choice of vertical elevation system, since the method must be reproducible for any Coordinate Reference System (CRS) and their respective vertical Datum. It is true that a city the size of São Paulo, which exceeds 47 km wide, will impact the calculated dimensions, a difference of just over 39 min of degrees. At that distance, for example, an object should be 400 m above the observer's altitude to be observable at this distance.[7]

4. Results

Visually in approximate scale, the results draw attention because they resemble the representation of an ambient occlusion with diffuse lighting, highlighting vertical elements such as buildings and their relationship with the immediate surroundings, as can be seen in the images on the left of Fig. 3. The Continuous Digital Model (CDM) of the SVF calculated for both cities (left column) has clear advantages in inferring the morphology of both cities more clearly than the SDM (middle) or the aerial photo (right). Comparing the images of Fig. 3 on the left with the one in the middle, it is possible to observe details of the shapes in a more emphatic way. Full and empty spaces gain nuances according to the proximity of a visual barrier, providing greater ability to differentiate more or less closed spaces. Although SVF is based on the relationship between the heights of the buildings, it can better translate the relation of the height of one building considering how neighboring buildings with different heights would affect sky view from a certain point of view. Fig. 3 depicts this comparison, where SVF-CDM brings a more detailed view than only analyzing building heights (DSM). The result is a more extraordinary richness of detail in a continuous digital model from the perspective of the terrain, from the pedestrian's view. Such a paradigm shift represents a breakthrough for the approach from the human perspective, something unprecedented for a continuous digital surface model that strongly contributes to relating urban form with human-centered studies.

The continuous digital model of SVF (SVF-CDM), resulting from SVFPy, allows the visual identification of patterns of urban morphology. In these places, which can be easily observed in Figs. 4 and 5, there is a clear distinction, either visually or statistically between each one of them, representing a breakthrough for studies from the urban form to the scale of the immediate surroundings of the building in the human perspective, the pedestrian. This advance opens up possibilities for visually and automatically distinguishing portions of similar urban fabrics in future works.

In the city of São Paulo, it is possible to identify distinct patterns of morphology without the need to overlap road axes, just by observing the SVF-CDM, as observed in Fig. 4. In this sample of 9 chosen locations, it can be stated that the city of São Paulo presents heterogeneity in its morphology at the same time, in which it finds homogeneous patterns within the limits established by visual criteria by the polygons. Highlights 5 and 6 show typologies of favelas that still have distinct characteristics. For example, in item 5, the alleys configure blocks with canyons that are access alleys, unlike item 6, which is more homogeneous, making each one distinct despite having similarities. The higher variance and the lower median in item 5 compared to 6 suggest that the canyons formed in item 5 are more marked than in item 6. Item 2 presents canyons much more prominent than the others and totally different from item 1, a horizontal residential neighborhood with a more generous road layout in relation to the heights of the buildings. This fact can still be identified by the statistical description. Items 3 and 4 are orthogonal housing patterns. The proximity of the

[6] https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm.

[7] Quick calculation considering the secant of the angle formed between two coordinates, considering the radius of the Earth at the Equator (WGS84) and disregarding the Geoidal model flattening.

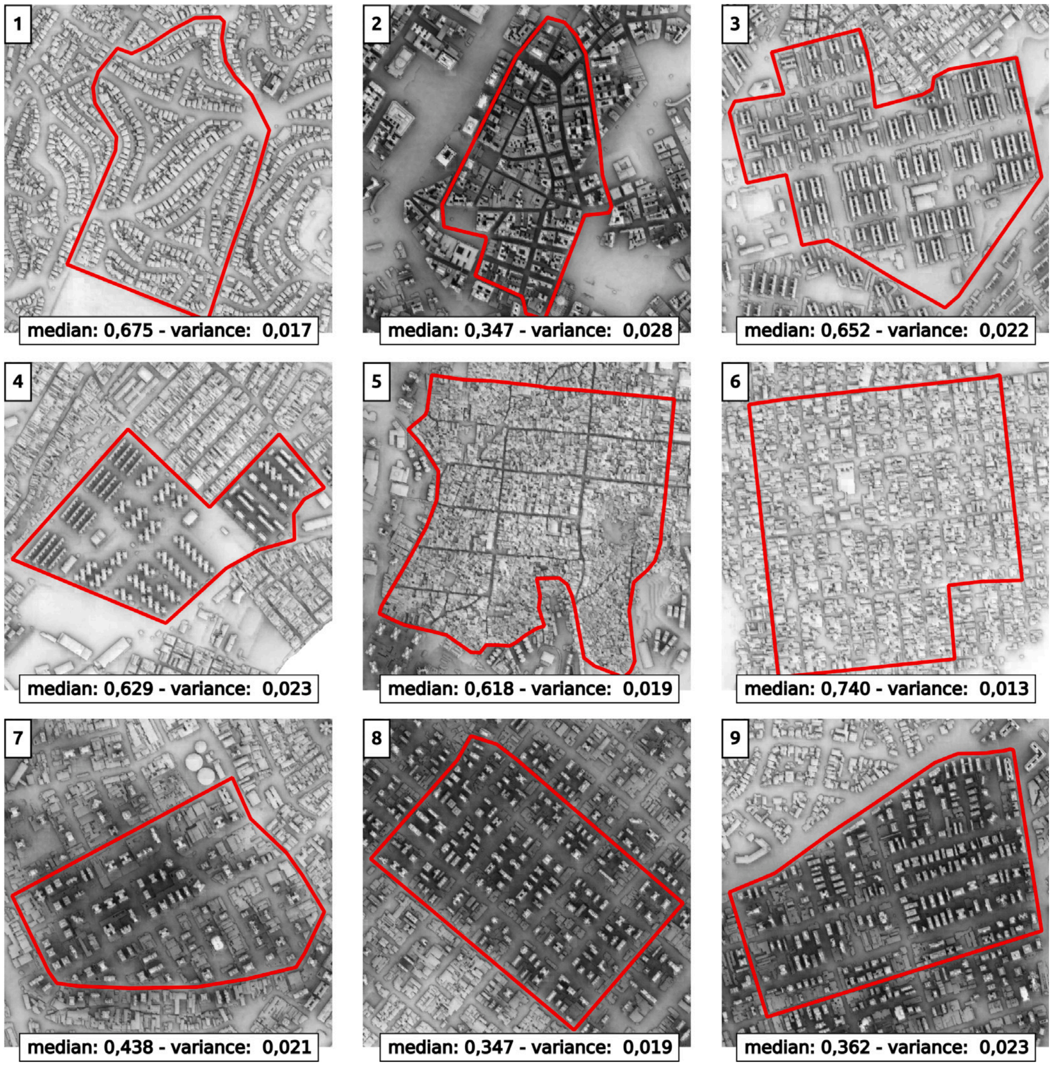

Fig. 4. Continuous digital model of Sky View Factor calculated for the city of São Paulo, highlighting portions of contiguous areas with morphological similarities, and median ($\tilde{x}$) and variance (σ^2) written down.

volumes suggests that occasional enclosures are interspersed with freer areas. However, the areas that contribute with the highest median, that is, with a greater view of the sky, are the tops of buildings, a private area. Differently, items 7, 8 and 9 present a vertical pattern of individual lots with lateral setbacks, promoting a more significant limitation of the view of the sky and making these 3 landscapes very similar both visually and statistically. The interposition of volumes and their effects on the immediate surroundings characterize each item in the last line, showing the capacity of the proposed model to characterize certain regions, even if they are very similar. Many of these insights have already been realized by pedestrians and desktop researchers in photo-based case studies, drone flights, and landscape studies. Still, such a procedure can now be largely performed on a digital model, representing a breakthrough for the study of free space systems and urban morphology.

For the city of New York, with some outstanding areas, the heterogeneity of urban morphology is even more evident from a succession of homogeneous morphological sets. For example, in items 1, 3 and 4 it is possible to clearly delineate the limits of the implementation of a certain homogeneous pattern of vertical housing, a legacy of modernism defended by Le Corbusier and criticized by Jacobs [44]. Despite close median values, variance distinctions suggest that each of these patterns has its peculiarities. Items 5 and 9 impress with the formation of canyons extremely designed to prevent sky visibility, as the buildings are tall and occupy the lot generously, without setbacks or distances. In these items, both the variance and the median have extreme values in the analyzed cases and suggest that statistically it is possible to intuit that values like this strongly suggest the formation of canyons. Item 6 also has well-marked canyons, with internal areas in the block with low sky view and has lower buildings that can be identified by the less dark tones between the blocks. Quite different from items 2, 7 and 8, which are more homogeneous, with grayer tones, median value close to 0.5, which means that the canyons are not as marked and deep because the buildings are lower than in the previous ones. However, for the latter, the characteristics of each one of them still stand out, either by the more reticulated layout of the streets, or by the pattern of occupation of more or less distant lots. This is very clear in the expression of individual variances. Although these studies have been carried out occasionally in some landscapes, even so, with an SVF-CDM it is possible to quantify and compare similar regions in a much broader way, expanding the discussion on urban form, even for a city widely studied in the literature like New York.

Comparing the highlights of the cities of São Paulo and New York, it is possible to state that there are much more impediments to seeing the sky in New York compared to São Paulo. With the exception of homogeneous housing patterns such as items 1, 3 and 4 of Fig. 5, in general, New York makes much more use of the space of lots with less side setbacks, making the landscape more continuous and uniform,

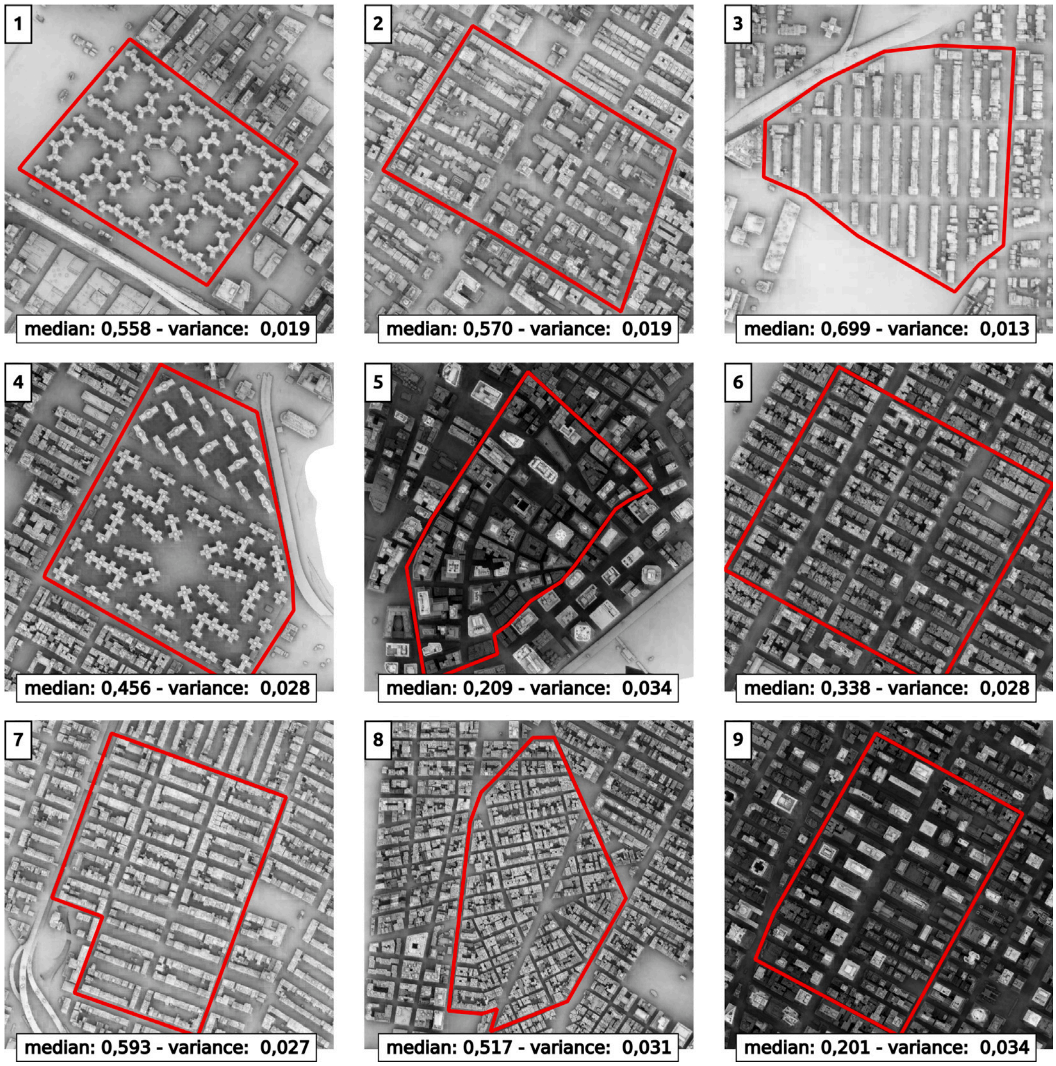

Fig. 5. Continuous digital model of Sky View Factor calculated for New York City, highlighting portions of contiguous areas with morphological similarities, and median ($\tilde{x}$) and variance (σ^2) written down.

which does not mean monotony, but diversity. In São Paulo, often as in items 7, 8 and 9 of Fig. 4 diversity does not prevent the formation of enclosures and a much more heterogeneous sky view pattern characterizing the landscape with isolated buildings in the lot center. That is, even for such different cities, the view of the sky is restricted as it is built and analyzing this becomes possible with the proposition of this methodology. It is not a comparative analysis of what is better or worse, but the SVF-CDM is a tool for the description and analysis of these landscapes and future landscapes, new buildings and planned volumes, which would not be possible in an extensive and continuous way . with the tools used so far. Especially when you want to compare the built form between cities, the SVF-CDM is an indicator that standardizes and can give reproducibility to studies on open space systems, especially considering the human perspective, something unprecedented for the entire urban fabric of a large city.

Therefore, the statistical behavior of the distribution of the SVF-CDM is an identifier of urban form by itself, regardless of the scale of study. This allows for comparison and analysis, as well as visual properties and the researcher's interpretive capacity, enabling its systematic use in mathematical and computational models. Using the same example items from Figs. 4 and 5 the same conclusions can be reached just by looking at some statistical components. For example, looking at median and variance of items 7, 8 and 9 of Fig. 4 it is possible to identify their similarities and nuances. Likewise, in Fig. 5, observing items 5, 6 and 9, their similar characteristics are observed without losing what characterizes each of these items individually. For future work, the SVF-CDM should be used as a segmentation dimension of morphological patterns using machine learning and artificial intelligence tools, which should contribute to more specific public policies for areas that have morphological similarity, whether for use and occupation of the soil, environmental policies or comfort on a human scale.

Even analyzing the entire data set, it is possible to identify, in the distribution of the calculated values of the SVF-CDM, indications that denote relevant morphological characteristics of each studied city, identifying its singularities. As shown in Fig. 3, New York City has a significant number of points with little view to the sky, characteristic of urban canyons. Unlike the city of São Paulo, mostly due to frontal setbacks, these formations have a different configuration. That is, there is a whole complexity expressed in a new calculated dimension that can be explored as evidence or index for studies, regulation and prescription of urban morphology even on a much larger scale as a city. For the city of São Paulo, the curve declines abruptly, after a sky view factor of 50%, which means a more sparse and heterogeneous verticalization. It is possible to observe in Fig. 6 that the visible sky ratio distribution is not uniform and is evident even on such a large scale. On the map, it is possible to associate a smaller proportion of the visible sky in the denser and more verticalized portions of the cities, however, with a heterogeneous distribution in space. In the city of São Paulo, the

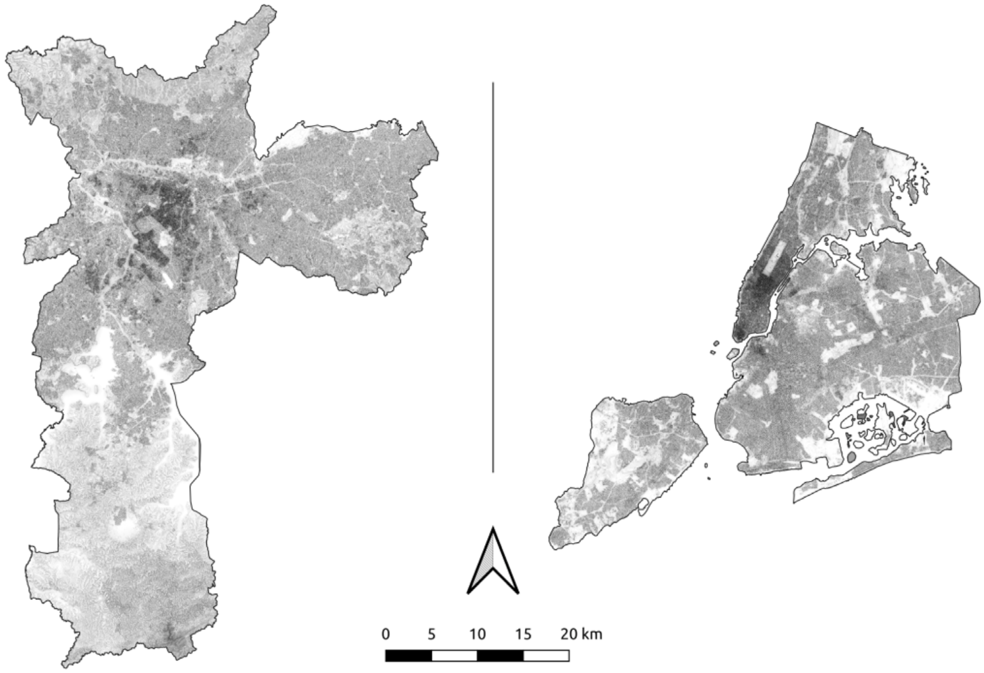

Fig. 6. SVF-CDM of São Paulo on the left and New York on the right.

"Marginals" stand out, largely due to their generous width which, despite representing a transposition barrier, still seems to preserve ample visibility of the sky, as well as in the areas of the reservoirs south of the city. Other points stand out, such as neighborhoods with low building density or restrictions on vertical development. For the city of New York, Manhattan stands out and its tall buildings that promote a darkened point in the model, with the exception of the highlight of Central Park. It is also evident that New York has a large area with a lower average SVF, which is compatible with the greater construction density, alternating with the green areas outside the island. Therefore, the SVF-MDL seems more effective than a traditional and monotonous model of urban expansion of the presence or absence of urbanization. It can identify nuances and aspects of urbanized form concentrated in just one dimension, which is critical for evidence-based public policy.

Studying the impact of land use indices, gauges and setbacks on the SVF in external areas and public roads should stimulate discussion about the impact that the built environment has on its surroundings, which is currently not something trivial to be done based on in evidence. The quality, use and occupation of the surroundings lack resources to quantify the amount of visible sky space, which can be understood as public and collective property, and is transferred inexpensively to those who decide or use their space without analyzing it. the impact on the neighborhood, verticalizing the buildings. Even in urban planning, there are few resources and metrics for discussions about the visual impact of neighborhoods, considering the complexity of form in the city. The theme of verticalization frequently runs through the city's science forums, especially in a scenario that encourages urban density without highlighting an indicator that measures its impact on public areas and open spaces. The SVF-CDM provides a synthetic way to express not only the aspect of verticalization, but also: the closure of roads by aligned building facades, the quality of lateral setbacks, excessive shading and the relationship with the vegetated environment. Such factors are essential, for example, to study walkability and discourage the use of individual private transport. Studying the urban form through the SVF-CDM gives rise, therefore, to a new perspective on the production of space in the city beyond the indices of the production of space in the lot, and thus starting from the consequence of the proposed building and its impact on the restriction of vision with the sky and in adjacent open space systems.

5. Final considerations and future works

The main purpose of this publication is to initiate a series of discussions on the use of SVF-CDM as an indicator of urban morphology and open space systems, in addition to the application of this method in the already established utilities described in the literature.

SVFPy is an unprecedented advance for geoprocessing and for urban sciences by presenting a computationally efficient method to calculate the sky view factor in large areas such as cities and even with the potential to create more models of the complexity of the built environment under the human perspective of the observer at ground level.

The SVFPy method was efficient to calculate the SVF for São Paulo and New York in a sub metric spatial resolution, but it has possibilities for improvements that should provide even better performance and more accuracy. However, the implementation still requires further adjusts to improve maintainability and code readability, in addition to adding functionality for GPU processing, which should follow the publication of this article and will result in new versions of SVFPy.

Future works should bring an exploratory and analytical methodology that can propose strategies for local and global scales, to validate its use as a descriptor of the urban landscape, as the ability to distinguish elements and heterogeneous groupings by methods of classification of the results. Or yet, bring evidence for other uses already discussed in the literature, such as the GNSS constellation sighting using GPX files or, at best, Rinex records of a sample of points in each of the cities studied.

The resulting data can easily generate other derived analytical dimensions such as, for example, number of hours of sunlight, dynamic or summarized shading, an area viewed per point, the 3D virtual dome of the enclosure, generation of a dynamic digital skyline (visible horizon profile), mapping or simulation of noise, sight at visual elements, among many others that depend on calculations from the perspective of the observer. This should be systematized in future works and used by other systems and methods.

An interesting finding in the São Paulo results that should motivate future work is the ability of the SVF-CDM to represent the relief without the bias of a single angle. Unlike conventional relief analyses, whose azimuthal angle is unique, the SVF-CDM calculation is carried out in multiple directions and at different scales, not restricted to nearby obstacles. Even so, the SVF-CDM combines the ability to represent slope and height since it represents the observer's relationship with the surroundings. Thus, the topography is entirely highlighted without the inconvenience of multiple interactions at different angles. Hence, a colateral finding of this study is that the SVF-CDM is an option to the list of tools for terrain analysis and should motivate future work, above all for those analyses that depend on the sight or applications that serve as a "proxy" for celestial aspects such as insolation, constellation sight, radio transmission, among many others. The topographic factor is relevant when comparing the city of São Paulo to New York, as it has a strong influence, even in the most urbanized environment, reinforcing the use of the SVF-CDM as a resource for describing the urban environment, independent of any other factor.

For both cities, the vegetation classes were disregarded for the creation of the DSM (Digital Surface Model), since unlike the buildings and the terrain, the vegetation partially impedes the view of the sky, and even though for many arboreal individuals, the degree of this impediment is seasonal. There are vegetation vision factor models that can be derived from the currently proposed method and that should encourage future work, such as the vegetation vision index and the vision index by classes.

With the aim of spreading and democratizing access to knowledge, this method is available in open source libraries to be used and adapted with autonomy and ease by other researchers and the processed data are shared in repositories to avoid the reprocessing of similar data and encourage the reproducibility of the methods presented here.

CRediT authorship contribution statement

Fernando Gomes: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Mariana Giannotti:** Writing – review & editing, Supervision.

Declaration of competing interest

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Data availability

Data will be made available on request.

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